ECON 770 Recitation 1

Aug 14, 2020

THE UNIVERSITY of NORTH CAROLINA at CHAPEL HILL

University of North Carolina at Chapel Hill

Introduction to Recitation

- ECON 770 recitation versus others
- This year versus last year
- Slides and homework
- Midterm, final and qual
- *All the parts with star (*) in slides are not required*

Office Hour

- Email: tianqili@ad.unc.edu
- Wednesday 1:00pm-2:00pm with Zoom ID *97741058246*
- Friday 11:30am-12:30pm with Zoom ID *93087202270* (as an extension to recitation)

Homework / Assignments

- Please make your homework legible, write down things you understand
- Discussion is allowed, but copy is prohibited
- Recommendation on homework:
  - Use LaTeX (if you are familiar with and if it won't take a long time)
  - Write on paper and scan them as a PDF
  - Write on electronic devices (e.g. iPad, Surface, etc) and export as a PDF

Course Contents

- Probability Theory (Definitions, PDF, CDF), Random Variables (Random Vector, Independence, Transformation, MGF), Distributions (Multi-dimensional Normal, Algebra, Asymptotic Property), Expectations (LIE, Moments)
- Statistics
  - Point and Interval Estimation
  - Estimator Properties (Unbiasedness, Consistency, Efficiency)
  - Hypothesis Testing (Finite, Infinite, Speed of Convergence)
  - Maximum Likelihood: Theory, Estimation, Inference (Specification, Consistency, Wald Test, LR Test, Score Test)

Definitions

Please make sure you know all the definitions,

- Experiment (e.g. roll a die, flip a coin), Outcome ω, Sample Space Ω
- Set, Element, Subset, Empty Set, Union, Intersection, Ven Diagram[1], Compliment, Disjoint Sets, Partition
- Countable, Uncountable, σ-field (or σ-algebra)
- Probability Function, Independence (Pairwise and Mutual), Conditional Probability

[1] Good to use when you analyze, but it cannot be a formal proof in exam.

De Morgan's Laws

Theorem

Suppose that $\{A_\lambda\}_{\lambda\in\Lambda}$ is a series of sets, then

$$\left(\bigcup_{\lambda\in\Lambda} A_\lambda\right)^c = \bigcap_{\lambda\in\Lambda} A_\lambda^c \text{ and } \left(\bigcap_{\lambda\in\Lambda} A_\lambda\right)^c = \bigcup_{\lambda\in\Lambda} A_\lambda^c$$

Proof.

(Partly) Suppose that $x \in \left(\bigcup_{\lambda\in\Lambda} A_\lambda\right)^c$, then $x \notin \bigcup_{\lambda\in\Lambda} A_\lambda$, which implies that $\forall\ \lambda \in \Lambda$, $x \notin A_\lambda$. This means $x \in A_\lambda^c$ for all λ. Thus, $x \in \bigcap_{\lambda\in\Lambda} A_\lambda^c$ and $\left(\bigcup_{\lambda\in\Lambda} A_\lambda\right)^c \subset \bigcap_{\lambda\in\Lambda} A_\lambda^c$. From the other side, suppose that $x \in \bigcap_{\lambda\in\Lambda} A_\lambda^c$, then for all λ, $x \in A_\lambda^c$, suggesting that $x \notin A_\lambda$. As a result, $x \notin \bigcup_{\lambda\in\Lambda} A_\lambda$. $\square$

σ-field

Defn.

Let $\mathcal{B}$ be a collection of subsets of set Ω, then it is a σ-field if

- $\varnothing \in \mathcal{B}$
- If $A \in \mathcal{B}$, then $A^c \in \mathcal{B}$
- If $A_i \in \mathcal{B}$ $(i = 1, 2, \dots)$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{B}$

- Example 1: $\{\varnothing, \Omega\}$ (smallest)
- Example 2: Power set $\mathcal{P}(\Omega)$ (biggest if Ω is discrete)
- Example 3*: Borel set - Any set in a topological space that can be formed from open sets (or, equivalently, from closed sets) through the operations of countable union, countable intersection, and relative complement (e.g. 1-dim Borel set $\mathcal{B} = \{(a, b] : \ \forall a, b \in \mathbb{R}\}$)

Exercise I*

Suppose that S is a non-empty set and $S \subset \Omega$, then obviously $S \in \mathcal{P}(X)$ where power set $\mathcal{P}(X)$ is a σ-field. Let

$$\mathcal{F} = \cap\{\mathcal{F}' : \mathcal{F}' \text{ is a } \sigma\text{-field s.t. } S \in \mathcal{F}'\}$$

Prove that

- $\mathcal{F}$ is a σ-field
- If $\mathcal{B}$ is a σ-field s.t. $S \in \mathcal{B}$, then $\mathcal{F} \subset \mathcal{B}$

Probability Function

Defn.

Probability Function is a mapping $P : \mathcal{B} \to [0, 1]$ satisfying three axioms

- $P(A) \geq 0$ for any event $A \in \mathcal{B}$
- $P(\Omega) = 1$
- If $A_i \in \mathcal{B}$ $(i = 1, 2, \dots)$ are disjoint event, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Notes:

- Probability function is defined on σ-field to make it complete
- A general measure λ is also defined on σ-field and satisfies $\lambda(A) \geq 0$

Question: what is *Prob*(randomly select a number and it's rational)?

Lebesgue Outer Measure*

Defn.

Given a subset $E \subset \mathbb{R}$, with the length of interval $I = (a, b)$ and $m(I) = b - a$, the Lebesgue outer measure $\lambda^*(E)$ is defined as

$$\lambda^*(E) = \inf \left\{ \sum_{k=1}^{\infty} m(I_k) : (I_k)_{k \in \mathbb{N}} \text{ are open intervals with } E \subseteq \bigcup_{k=1}^{\infty} I_k \right\}$$

Then we can show that for any countable set $A \subset \mathbb{R}$, we have $\lambda^*(A) = 0$. Suppose $A = \{a_1, a_2, \dots\}$ is a countable set, then $\forall\ \varepsilon > 0$, we know

$$\bigcup_{k=1}^{\infty} \left(a_k - \frac{\varepsilon}{2^{k+1}}, a_k + \frac{\varepsilon}{2^{k+1}} \right)$$

fully cover A, then

$$0 \leq \lambda^*(A) \leq \sum_{k=1}^{\infty} m\left(\left(a_k - \frac{\varepsilon}{2^{k+1}}, a_k + \frac{\varepsilon}{2^{k+1}} \right)\right) = \sum_{k=1}^{\infty} \frac{\varepsilon}{2^k} = \varepsilon \ (\to 0)$$

Propositions

Theorem

Given sample space Ω, σ-field $\mathcal{B}$ and probability function P, we have, $\forall A, B \in \mathcal{B}$

- $P(A^c) = 1 - P(A)$
- $P(\varnothing) = 0$
- $P(A) \leq 1$
- If $A \subseteq B$, then $P(A) \leq P(B)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B) \leq P(A) + P(B)$
- $P(A \cap B) \leq P(A) + P(B)$
- For $A_i \in \mathcal{B}$ $(i = 1, 2, \dots)$, $P(\cap_{i=1}^{\infty} A_i) \leq \liminf_{i \to \infty} P(A_i)$

Lower Limit and Upper Limit of Sequences

Defn.

$$\varliminf_{n\to\infty} x_n \equiv \liminf_{n\to\infty} x_n := \lim_{n\to\infty}\left(\inf_{m\geq n} x_m\right)$$

$$\varlimsup_{n\to\infty} x_n \equiv \limsup_{n\to\infty} x_n := \lim_{n\to\infty}\left(\sup_{m\geq n} x_m\right)$$

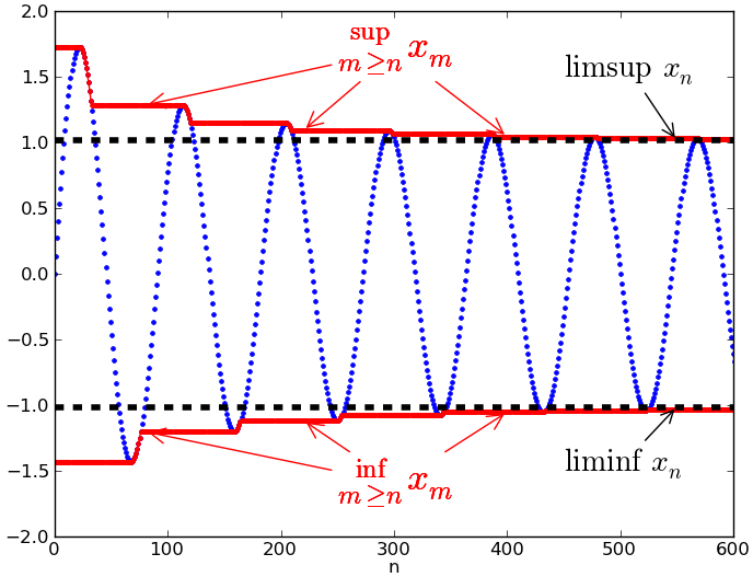

Figure: From wikipedia

Lower Limit and Upper Limit of Sets

Defn.

$$\liminf_{n\to\infty} X_n = \sup_{n\geq 1}\left\{\inf_{m\geq n}\{X_m\}\right\} = \bigcup_{n=1}^{\infty}\left(\bigcap_{m=n}^{\infty} X_m\right)$$

$$\limsup_{n\to\infty} X_n = \inf_{n\geq 1}\left\{\sup_{m\geq n}\{X_m\}\right\} = \bigcap_{n=1}^{\infty}\left(\bigcup_{m=n}^{\infty} X_m\right)$$

Example:

$$\{X_n\} = \{\{0\},\{1\},\{0\},\{1\},\{0\},\{1\},\ldots\}$$

In this case,

$$\liminf_{n\to\infty} X_n = \varnothing \text{ and } \limsup_{n\to\infty} X_n = \{0,1\}$$

Question: What's the relationship between

$$\bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty} A_k \quad \text{and} \quad \forall n\in\mathbb{N}_+,\ \exists k\geq n \text{ s.t. } A_k$$

Exercise II*

Show that

- If $\{A_n\}$ is monotonically increasing[2], then

$$\lim_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} A_n$$

- If $\{A_n\}$ is monotonically decreasing[3], then

$$\lim_{n\to\infty} A_n = \bigcap_{n=1}^{\infty} A_n$$

(Hint: You may use $\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k$ or $\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$ to show the existence of $\lim_{n\to\infty} A_n$)

[2] For any $k \geq 1$, $A_k \subset A_{k+1}$

[3] For any $k \geq 1$, $A_{k+1} \subset A_k$

Independence

Defn.

- Events A and B are independent if and only if

$$P(A \cap B) = P(A)P(B)$$

- Events A_i $(i = 1, ...n)$ are pairwise independent if for **any** i and j,

$$P(A_i \cap A_j) = P(A_i)P(A_j)$$

- Events A_i $(i = 1, ...n)$ are mutually independent if every event is independent of any intersection of the other events, i.e. $\forall k \leq n$,

$$P(\bigcap_{j=1}^{k} A_{i_j}) = \prod_{j=1}^{k} P\left(A_{i_j}\right)$$

holds for all $\{i_1, \ldots, i_k\} \subseteq \{1, 2, \ldots, n\}$

Independence (Continued)

Here's an example for mutually independence. Suppose that we have four events A, B, C and D to be mutually independent, then we need to show

$$P(A \cap B) = P(A)P(B), P(A \cap C) = P(A)P(C), P(A \cap D) = P(A)P(D),$$
$$P(B \cap C) = P(B)P(C), P(B \cap D) = P(B)P(D), P(C \cap D) = P(C)P(D),$$

$$\begin{aligned} P(A \cap B \cap C) &= P(A)P(B)P(C), \\ P(A \cap B \cap D) &= P(A)P(B)P(D), \\ P(A \cap C \cap D) &= P(A)P(C)P(D), \\ P(B \cap C \cap D) &= P(B)P(C)P(D), \end{aligned}$$

and

$$P(A \cap B \cap C \cap D) = P(A)P(B)P(C)P(D)$$

Conditional Probability

Defn.

For events A and B in σ-field $\mathcal{B}$, conditional probability is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

where $P(B) > 0$

- $P(A|\Omega) = P(A)$
- $P(A|B) = P(A)$ if A and B are independent
- $P(A \cap B|C) = \frac{P(A \cap B \cap C)}{P(C)} = \frac{P(A \cap C|B)P(B)}{P(C)}$
- **Bayes' Rule**
- Good reference:

  `https://en.wikipedia.org/wiki/Conditional_probability`

Exercise III

(Casella and Berger, 1.6) Two pennies, one with $P(head) = u$ and one with $P(head) = w$, are to be tossed together independently. Define

$$p_0 = P(0 \text{ heads occur}),$$
$$p_1 = P(1 \text{ head occurs}),$$
$$p_2 = P(2 \text{ heads occur}).$$

Can u and w be chosen such that $p_0 = p_1 = p_2$? Prove your answer.

Exercise IV

(Casella and Berger, 1.11) Let S be a sample space.

(a) Show that the collection $\mathcal{B} = \{\varnothing, S\}$ is a sigma algebra.

(b) Let $\mathcal{B} = \{\text{all subsets of } S\text{, including } S \text{ it self}\}$. Show that $\mathcal{B}$ is a sigma algebra.

(c) Show that the intersection of two sigma algebras is a sigma algebra.